

GRID Prototype Report

DER targeting, campaign intelligence, and prequalified open-bid execution for Dominion Energy

SJ_Jain_Systems

2026-05-27

1 Executive summary of the prototype

GRID is a prototype decision tool for Dominion Energy. It connects three pieces that are usually treated separately: household targeting, DER campaign design, and the utility business case behind DER adoption.

The current prototype has three working surfaces:

1. **Targeting Explorer** ranks households by adoption priority, affordability impact, grid flexibility, and customer readiness.
2. **Map + Campaign Manager** turns the household scores into county-level campaign plans, recommended offers, and launch timing.
3. **Dominion Savings** estimates avoided infrastructure cost, DER grid value, marketing efficiency, subsidy value, virtual power plant value, emergency capacity value, and opportunity cost.

The first part of the tool answers the customer question: which households are good candidates, what is stopping them, and what offer would make adoption more likely? The second part answers the utility question: does the adoption create enough peak reduction, affordability value, or avoided infrastructure cost to matter for Dominion?

The next step is more operational. Once Dominion knows where DER programs or reliability investments make sense, the remaining problem is getting the work done. That is where the project should move toward a contractor-bidding layer.

The future idea should be framed narrowly. This is not a proposal to turn utility infrastructure into a public marketplace or to award critical work to the lowest bidder. The better claim is that some DER-related installation work is repeatable enough to be standardized. If Dominion can define the scope clearly and limit bidding to qualified vendors, then those jobs can be compared more like commodities without treating safety, reliability, or customer trust as commodities.

The central behavior problem is cost and trust. Customers hesitate when installation feels expensive, confusing, or risky. Utilities hesitate when contractors lack credibility, safety history, insurance, or a clear performance record. A controlled open-bidding system can address both sides: price competition pushes installation costs down, while prequalification and contractor scoring protect trust.

The prototype therefore builds toward this thesis:

GRID should identify high-value DER opportunities, translate them into standardized work packages, and help Dominion compare bids from prequalified contractors using both cost and trust metrics.

In this framing, the product is not just a marketing dashboard. It is a bridge from analysis to execution. The household model finds the opportunity. The savings model estimates why the opportunity matters. The future bidding layer helps Dominion decide who can deliver the work safely, quickly, and at a reasonable cost.

Question	Prototype output
Which households should Dominion prioritize?	Household ranking and DER pathway
Why might this household not convert?	Dominant barrier score and explanation
What should Dominion offer?	Financing, incentive, campaign message, and intervention
Why does adoption matter to the utility?	Peak reduction, avoided infrastructure cost, VPP value, and net savings
How can Dominion execute the work?	Future prequalified bidding layer for standardized DER work

The careful assumption is that the grid itself is not a commodity. However, parts of the installation layer around DER adoption can be made more commodity-like when the work is standardized, vendors are screened in advance, and bids are judged by risk-adjusted value rather than price alone.

1.1 Our Current Prototype

Visit the prototype at: [GRID for Dominion Energy](#)

2 What the prototype currently does

The website is organized around three tabs, each tied to a specific decision.

2.1 Targeting Explorer

The Targeting Explorer is the front door of the prototype. It allows a utility analyst to filter households by county, income band, DER type, targeting strategy, minimum adoption score, and outreach goal.

The output is a ranked table of the top households. Each row includes:

- Household ID and county.
- Priority score.
- Recommended DER technology.
- Recommended message.
- Financing option.
- Incentive type.
- Contact timing.
- Top adoption barrier.
- Barrier score.
- Dominion offer.

This tab keeps the model explainable. Dominion would not only see that a household ranks highly; it would see why that household is a good target and what intervention should be paired with it.

2.2 Map + Campaign Manager

The second tab converts household-level model outputs into action. It contains two pieces:

Component	Purpose
County DER Initiatives Map	Shows where priority households cluster and what DER initiative fits each county
Campaign Manager	Groups households into campaign segments based on barrier, technology, and objective

The Campaign Manager supports multiple objectives:

- Maximize adoption.
- Maximize affordability impact.
- Maximize grid flexibility.
- Maximize customer readiness.

This matters because the “best” household depends on Dominion’s goal. A household that is most likely to adopt may not be the same household that creates the highest grid value or the same household that most needs affordability support.

2.3 Dominion Savings tab

The Dominion Savings tab expands the prototype from customer targeting into utility economics. It estimates:

- Peak load reduction.
- Avoided infrastructure cost.
- Incentive/program cost.
- Net Dominion savings.
- Return on investment.
- Virtual power plant value.
- Emergency capacity value.
- Marketing efficiency.

- Subsidy value.
- Opportunity-cost savings.

The key insight is that DER adoption can be modeled as an alternative to infrastructure buildout. Utilities often build around peak-load, reliability, and redundancy constraints. If targeted DER adoption can reduce those constraints, GRID can estimate how much infrastructure spending may be delayed, avoided, or better prioritized.

This is where the prototype starts to point beyond campaign design. If the system can identify where adoption, upgrades, or reliability investments should happen, Dominion still has to coordinate the people who perform the work. That may include line contractors, engineering support, installation partners, surveyors, vegetation crews, right-of-way support, and DER-related vendors. A future version of GRID should help package this work for prequalified bidding. The point is not to make critical grid work a lowest-price auction. The point is to make repeatable scopes easier to define, compare, bid, award, and evaluate after completion.

Area	Question	Output
Targeting Explorer	Who should be prioritized?	Ranked households and offers
Map and Campaign Manager	How do scores become campaigns?	County guidance and launch timing
Dominion Savings	What is the business case?	Avoided cost, VPP value, and ROI

3 Prototype data and product logic

3.1 Data used in the prototype

The prototype uses a Virginia household sample. Each household includes demographic, housing, energy, engagement, DER-readiness, installer-market, rebate, and affordability features. To keep the PDF readable, the table below uses a compact summary rather than the full wide descriptive table.

feature	mean	median	min	max
income	107166	102672	35127	209845
energy Burden	4.69	4.6	1.2	12.2
peak57	8.8	8.8	4	13.8
home Age	29.92	30	5	75
outage Risk	0.34	0.33	0.12	0.72
engagement	0.55	0.55	0.2	0.9
estimated Install Cost	18022.8	17846.5	12270	25249
financing Acceptance	0.58	0.58	0	1
solar Suitability	0.52	0.52	0	1
roof Suitability	0.35	0.36	0.05	0.84
installer Count25mi	57.49	54	27	113
required Step Count	7.15	7	4	11
rebate Count	3.01	3	0	5
auto Matched Rebate Count	1.77	2	0	5

The model uses variables such as:

Variable	Meaning in the prototype
income	Household income proxy
energyBurden	Utility bill burden as a share of household income
peak57	5-7 PM peak demand proxy
homeAge	Age of the home
ev	EV ownership indicator
smartThermostat	Existing smart thermostat indicator
outageRisk	Reliability/resilience exposure proxy

Variable	Meaning in the prototype
engagement	Customer engagement score
financingAcceptance	Likelihood of accepting financing
solarSuitability	Solar potential score
roofSuitability	Roof feasibility score
installerCount25mi	Installer access within 25 miles
installerAvgRating	Installer trust/quality proxy
requiredStepCount	Program complexity proxy
rebateCount	Number of possible rebates
autoMatchedRebateCount	Number of rebates automatically matched
savingsEstimateConfidence	Confidence in the household savings estimate
publicChargerDistance	EV charging-access proxy
garageOrDriveway	Home charging feasibility proxy
panelCapacityProxy	Electrical panel readiness proxy
estimatedInstallCost	Estimated installation-cost burden

3.2 Prototype logic map

GRID follows a five-step product pipeline in the current prototype:

Raw household features → normalized model features
→ adoption, affordability, grid, and readiness scores
→ barrier diagnosis
→ campaign and DER recommendation
→ utility-value simulation.

This matters because prediction is not the final answer. It is the starting point. The system also explains the bottleneck, recommends the intervention, and estimates the utility-level impact.

The next version should add one more step:

Utility-value simulation → work package definition
→ open bid release
→ contractor scoring
→ award recommendation and execution tracking.

This addition changes the narrative. The current prototype starts with households. The next version should end with execution. Dominion would know which customers or circuits are worth targeting, what work needs to be performed, which vendors are qualified for that type of work, and how each bid compares on cost, timing, safety, reliability, and credibility.

3.3 Strategic narrative: from DER targeting to open-bid execution

The interview behind this project made one point especially clear: utilities are cautious for a reason. They do not hand work to the newest vendor just because that vendor is cheaper or has a cleaner pitch. They care about safety, reliability, insurance, proven experience, regulatory exposure, and whether the contractor can actually perform under utility standards.

That makes the bidding idea stronger, but only if it is framed correctly. GRID is not claiming that utility infrastructure is a commodity. It is claiming that some DER installation and support tasks can become more comparable once Dominion standardizes the scope and only allows qualified vendors to bid.

The future platform should support Dominion's existing teams rather than replace them. It would make contractor discovery, bid comparison, and vendor evaluation easier for the people already responsible for infrastructure projects and customer programs.

Utility concern	Why it matters	Future GRID response
Reliability	Poor execution affects customers and public trust	Score vendors by completion history, outage impact, and quality record
Safety	Utility work carries high liability	Require safety metrics, incident history, insurance, and compliance documents
Credibility	Utilities hesitate to use unknown vendors	Build verified vendor profiles and qualification tiers
Cost control	Installation work can become expensive at scale	Compare bids against expected cost ranges and avoided-cost value
Geography	Contractors must serve the actual territory	Match vendors to counties, circuits, travel radius, and crew availability
Timing	Peak seasons and storm restoration affect capacity	Include availability, mobilization speed, and schedule risk
Regulatory defense	Spending needs to be explainable	Preserve an audit trail for why a vendor was shortlisted or selected

This makes open bidding a practical future direction. If Dominion wants to expand DER adoption, manage electrification, handle data-center load growth, improve reliability, and avoid unnecessary infrastructure buildout, knowing what to do is only half of the problem. The other half is making the work easier to contract without weakening safety or reliability standards.

3.4 Careful commodity framing

The cleanest way to explain the idea is partial commoditization. The grid is not a commodity. Large utility infrastructure is too risky, regulated, and reliability-sensitive to be treated that way. But some DER installation tasks are repetitive enough to be standardized.

Examples include EV charger installation, smart thermostat installation, panel-readiness checks, solar assessment, weatherization packages, and other behind-the-meter support work. These jobs still require trust, but the scope can often be written in a comparable format.

This gives GRID a defensible procurement thesis:

Dominion should not commoditize the grid. Dominion should make the repeatable DER installation layer more comparable, more transparent, and easier to bid.

That distinction matters. A lowest-bid auction would be risky because it could reward underbidding, poor workmanship, or unsafe execution. A prequalified bidding system is different. It creates price pressure only after Dominion has screened contractors for safety, insurance, experience, geography, and past performance.

The two behavioral barriers are cost and trust:

Barrier	Why it slows DER deployment	Open-bidding response
Cost	High installation costs weaken customer adoption and the utility business case	Create competition among qualified contractors and compare bids against expected cost ranges
Trust	Customers and utilities hesitate when installer quality is unclear	Use prequalification, verified profiles, safety records, customer feedback, and performance history

This is the bridge between the prototype and the future platform. The prototype identifies where DER deployment has value. The bidding layer asks how Dominion can deliver that value at a lower risk-adjusted cost.

4 Mathematical methodology

This section formalizes the prototype as a scoring and simulation system.

4.1 Notation

All variables below are written in LaTeX inline math so they render cleanly in Quarto.

- i = household index.
- x_i = feature vector for household i .
- N = number of targeted households.
- p_i = baseline probability that household i adopts a DER.
- $\Delta p_i(u)$ = conversion lift from intervention u .
- r_i = expected peak reduction in kW if household i adopts.
- $c_i(u)$ = campaign and incentive cost for household i under intervention u .
- v^{cap} = value per kW of avoided peak capacity.
- v^{VPP} = value per kW of flexible VPP capacity.
- B_{ik} = severity of adoption barrier k for household i .
- S_i = overall household priority score.
- U_i = expected utility value created by household i .

4.2 Normalization

Most prototype variables are transformed onto a 0-to-1 scale before scoring. For a feature x_i , min-max normalization is:

$$\tilde{x}_i = \frac{x_i - \min_j(x_j)}{\max_j(x_j) - \min_j(x_j)}$$

When a higher raw value is worse, the normalized score can be reversed:

$$\tilde{x}_i^{\text{rev}} = 1 - \tilde{x}_i$$

The report uses $\text{clip}(z, 0, 1)$ to keep all intermediate scores between 0 and 1:

$$\text{clip}(z, 0, 1) = \begin{cases} 0, & z < 0 \\ z, & 0 \leq z \leq 1 \\ 1, & z > 1 \end{cases}$$

4.3 Adoption probability model

The prototype uses an interpretable weighted score rather than a black-box model. A deployable version could replace this with logistic regression, gradient boosting, or calibrated machine learning, but the current version keeps the math explainable.

$$p_i = \text{clip} \left(0.20\tilde{I}_i + 0.20\tilde{P}_i + 0.15E_i + 0.10V_i + 0.10F_i + 0.15R_i + 0.10T_i, 0, 1 \right)$$

Where:

- $\tilde{I}_i$ = normalized income/readiness proxy.
- $\tilde{P}_i$ = normalized 5-7 PM peak demand.
- E_i = engagement score.
- V_i = EV ownership or EV-readiness signal.

- F_i = financing acceptance.
- R_i = DER suitability score.
- T_i = smart thermostat or automation-readiness signal.

4.4 Barrier diagnosis model

GRID's main advantage is that it does not only estimate adoption probability. It also estimates why a household may not convert. Each barrier is scored from 0 to 100.

General form:

$$B_{ik} = 100 \cdot \text{clip} \left(\sum_{m=1}^{M_k} w_{km} \tilde{x}_{im}, 0, 1 \right)$$

The dominant barrier is:

$$k_i^* = \arg \max_k B_{ik}$$

The recommended Dominion intervention is then selected as:

$$u_i^* = g(k_i^*)$$

where $g(\cdot)$ maps a diagnosed barrier to an intervention such as on-bill financing, automatic rebate matching, a vetted installer marketplace, a switching concierge, or an income-qualified upgrade package.

4.5 Campaign segmentation model

The Campaign Manager groups households by dominant barrier and DER pathway:

$$\text{segment}(i) = (k_i^*, d_i)$$

where d_i is the recommended DER technology for household i .

For campaign segment s , the audience size is:

$$A_s = \sum_{i=1}^N \mathbf{1}\{\text{segment}(i) = s\}$$

The average barrier severity is:

$$\bar{B}_s = \frac{1}{A_s} \sum_{i:\text{segment}(i)=s} B_{ik_i^*}$$

Expected adoption lift for segment s is modeled as:

$$\Delta_s = \text{clip} \left(0.08 + \alpha \bar{S}_s + \beta \frac{\bar{B}_s}{100} + \gamma \log_{10}(A_s + 1), 0.03, 0.45 \right)$$

4.6 Utility-value model

Expected adopters are calculated as baseline adopters plus campaign lift:

$$\widehat{A} = \sum_{i=1}^N p_i (1 + \Delta_i)$$

Expected peak reduction is:

$$\widehat{R} = \sum_{i=1}^N p_i (1 + \Delta_i) r_i$$

Avoided infrastructure value is:

$$V^{\text{infra}} = \widehat{R} \cdot v^{\text{cap}}$$

Virtual power plant value is:

$$V^{\text{vpp}} = \widehat{R} \cdot v^{\text{vpp}}$$

Program cost is:

$$C^{\text{program}} = \sum_{i=1}^N p_i (1 + \Delta_i) c_i$$

Net utility value is:

$$V^{\text{net}} = V^{\text{infra}} + V^{\text{vpp}} + V^{\text{emergency}} + V^{\text{subsidy}} - C^{\text{marketing}} - C^{\text{program}}$$

5 Scoring formulas

5.1 Adoption priority score

The adoption priority score combines adoption probability, affordability value, grid value, incentive efficiency, and friction.

$$S_i^{\text{adopt}} = \frac{p_i \cdot A_i \cdot G_i}{C_i^{\text{incentive}}} \cdot (1 - F_i^{\text{friction}})$$

where:

- p_i = adoption probability.
- A_i = affordability value.
- G_i = grid value.
- $C_i^{\text{incentive}}$ = incentive cost.
- F_i^{friction} = adoption friction.

5.2 Affordability score

The affordability score prioritizes households where the same bill reduction creates a larger welfare improvement.

$$S_i^{\text{afford}} = 100 \cdot \text{clip} \left(0.45\tilde{B}_i + 0.25\tilde{P}_i + 0.20(1 - \tilde{I}_i) + 0.10\tilde{C}_i^{\text{install}}, 0, 1 \right)$$

where:

- $\tilde{B}_i$ = normalized energy burden.
- $\tilde{P}_i$ = normalized peak usage.
- $\tilde{I}_i$ = normalized income.
- $\tilde{C}_i^{\text{install}}$ = normalized installation cost.

5.3 Grid flexibility score

The grid flexibility score prioritizes households that can reduce system stress during peak windows.

$$S_i^{\text{grid}} = 100 \cdot \text{clip} \left(0.40\tilde{P}_i + 0.20O_i + 0.15EV_i + 0.15D_i + 0.10Q_i, 0, 1 \right)$$

where:

- O_i = outage risk.
- EV_i = EV ownership indicator.
- D_i = dispatchability of the recommended DER.
- Q_i = equipment or home-suitability readiness.

5.4 Customer readiness score

The readiness score estimates whether the household is prepared to act soon.

$$S_i^{\text{ready}} = 100 \cdot \text{clip} (0.35E_i + 0.25F_i + 0.20(1 - B_i^{\text{max}}/100) + 0.20H_i, 0, 1)$$

where:

- E_i = engagement.
- F_i = financing acceptance.
- B_i^{max} = highest barrier score.
- H_i = prior program history or eligibility signal.

5.5 Example barrier formulas

Upfront cost friction:

$$B_i^{\text{cost}} = 100 \cdot \text{clip} \left(0.34(1 - F_i) + 0.24\tilde{I}_i^{\text{pressure}} + 0.22\tilde{C}_i^{\text{install}} + 0.12\tilde{H}_i^{\text{lowvalue}} + 0.08R_i, 0, 1 \right)$$

Decision fatigue:

$$B_i^{\text{fatigue}} = 100 \cdot \text{clip} \left(0.42\tilde{S}_i^{\text{steps}} + 0.24\tilde{U}_i^{\text{complex}} + 0.18\tilde{P}_i^{\text{programs}} + 0.16\tilde{H}_i^{\text{history}}, 0, 1 \right)$$

Rebate confusion:

$$B_i^{\text{rebate}} = 100 \cdot \text{clip} \left(0.45G_i^{\text{automation}} + 0.30\tilde{R}_i^{\text{count}} + 0.15\tilde{S}_i^{\text{steps}} + 0.10N_i^{\text{new}}, 0, 1 \right)$$

Low ROI clarity:

$$B_i^{\text{roi}} = 100 \cdot \text{clip} \left(0.50(1 - K_i) + 0.22Q_i^{\text{gap}} + 0.16P_i^{\text{panelgap}} + 0.12H_i^{\text{age}}, 0, 1 \right)$$

High energy burden:

$$B_i^{\text{burden}} = 100 \cdot \text{clip} \left(0.75\tilde{B}_i^{\text{high}} + 0.15\tilde{I}_i^{\text{pressure}} + 0.10\tilde{P}_i, 0, 1 \right)$$

5.6 Infrastructure cost curve

GRID compares linear, stepwise, and stress-curve infrastructure cost assumptions.

Linear cost:

$$C^{\text{linear}}(R) = \lambda R$$

Stepwise cost:

$$C^{\text{step}}(R) = \sum_{m=1}^M K_m \mathbf{1}\{R > \tau_m\} + \eta \max(0, R - \tau_M)$$

Stress-curve cost:

$$C^{\text{stress}}(R) = R \cdot b \left(1 + \left(\frac{R}{R_0} \right)^\theta \right)$$

Opportunity-cost savings are:

$$\text{OpportunitySavings} = C^{\text{infra}}(R) - C^{\text{DER}}$$

6 Use cases and validation

6.1 Dominion Energy use cases

Use case	How GRID helps
Program targeting	Finds households most likely to respond to specific DER offers
Affordability strategy	Prioritizes high-energy-burden households for support
Peak reduction	Targets households with high 5-7 PM load and flexible DER pathways
Campaign planning	Builds barrier-specific campaign segments
Infrastructure planning	Estimates avoided peak-capacity investment
VPP strategy	Identifies customers who can become dispatchable grid assets
Contractor procurement	Future layer translates priority work into open bids for qualified vendors

Use case	How GRID helps
Vendor evaluation	Future layer scores contractors by safety, cost, credibility, geography, and timing
Regulatory reporting	Links DER programs to affordability, reliability, resilience, and procurement decisions

6.2 Customer use cases

GRID can also improve the customer experience. Instead of sending every customer the same generic program page, Dominion could provide:

- A personalized DER recommendation.
- A plain-language savings estimate.
- A financing option matched to affordability barriers.
- Automatic rebate matching.
- Installer trust support.
- A rate-plan recommendation.
- A low-friction enrollment pathway.

6.3 Validation plan

The prototype should eventually be validated against historical Dominion program data. A deployable version would use observed labels such as:

- Installed DER.
- Clicked or opened DER outreach.
- Started but abandoned rebate application.
- Contacted Dominion about solar, EV charging, thermostat, or weatherization.
- Enrolled in demand response.
- Enrolled in off-peak EV charging.
- Did not engage.

The main validation metrics should be:

$$\text{ConversionLift} = \frac{\text{conversion rate under GRID targeting}}{\text{conversion rate under broad outreach}}$$

$$\text{CostPerConversion} = \frac{C^{\text{marketing}} + C^{\text{program}}}{\text{number of conversions}}$$

$$\text{PeakKWReduced} = \sum_{i \in \text{adopters}} r_i$$

$$\text{InfrastructureROI} = \frac{V^{\text{infra}} - C^{\text{DER}}}{C^{\text{DER}}}$$

A pilot should compare a GRID-targeted campaign against a broad campaign or business-as-usual campaign using A/B testing.

7 Future solutions

The current version of GRID proves the first half of the product: Dominion can combine customer attributes, DER fit, barrier scoring, campaign design, and utility-value calculations in one decision tool. The next version should focus on the missing half: execution.

The future idea is a prequalified contractor-bidding layer for standardized DER work. It should help Dominion bid down installation and support costs, but only within a controlled system. The platform should not be framed as open outsourcing for all utility work. It should be framed as a procurement tool for repeatable scopes where comparison across vendors is possible.

The logic is straightforward:

Targeting insight → planned work → standardized scope → prequalified bids → completed deployment

This matters because a utility does not create value by identifying a good opportunity. It creates value when the opportunity becomes completed work. A model can show that a household is a strong EV charger candidate or that a county has high DER value, but Dominion still needs crews, installers, project coordination, and quality control.

The future system should make bidding easier without making it reckless. Contractors would compete, but only after meeting Dominion’s minimum standards. The cheapest bid would not automatically win. The best bid would be the one with the strongest risk-adjusted value.

7.1 Why open bidding fits the GRID narrative

The project started with DER adoption. The obvious problem was customer conversion: people may not adopt because of cost, rebate confusion, uncertain savings, low trust in installers, or decision fatigue. The prototype addresses that by scoring households and recommending offers.

The interview added another constraint. Even if the customer side is solved, utilities still have to execute the work. Contractors are not interchangeable. Utilities care about credibility, safety, insurance, relationships, and whether a vendor has performed similar work before. That means execution is also a trust problem.

Open bidding fits because it addresses the two main frictions at the same time:

Friction	Current problem	Platform response
Cost	Installation and support costs can make DER programs harder to justify	Qualified contractors compete on price and schedule
Trust	Utilities and customers hesitate when installer quality is uncertain	Vendors are screened, scored, and tracked over time

The prototype is therefore building toward a market-design problem: how can Dominion turn a stream of DER opportunities into clear work packages that trusted contractors can compete to deliver?

7.2 Future product: Dominion Open Bid

The future product could be called **Dominion Open Bid** or **GRID Bid**. It would be a structured procurement layer where Dominion publishes work packages and qualified contractors submit bids.

Commoditization should be defined carefully here. It does not mean the work is simple or risk-free. It means the parts that can be standardized should be standardized: scope, required documents, eligibility rules, bid fields, completion metrics, and post-project evaluation.

A work package could include:

- Location and service territory.
- DER type or project category.
- Required certifications.
- Insurance and safety requirements.
- Estimated project size.
- Completion window.
- Customer-contact expectations.

- Documentation required after completion.
- Expected utility value or avoided-cost rationale.

A contractor profile could include:

- Service territory.
- Crew availability.
- Certifications.
- Insurance status.
- Safety record.
- Past completion rate.
- Customer satisfaction.
- Average bid accuracy.
- Rework history.
- Experience with similar DER projects.

7.3 Bid scoring model

The future platform should score bids transparently. Price should matter, but it should not dominate the decision. A cheaper bid can become expensive if it causes delays, rework, customer complaints, or safety problems.

A possible score is:

$$S_{j,w}^{\text{bid}} = \alpha C_{j,w} + \beta R_j + \gamma T_{j,w} + \delta G_{j,w} + \eta Q_j$$

where:

- $S_{j,w}^{\text{bid}}$ = bid score for contractor j on work package w .
- $C_{j,w}$ = cost competitiveness.
- R_j = reliability and safety record.
- $T_{j,w}$ = timeline fit.
- $G_{j,w}$ = geographic fit.
- Q_j = qualification and credibility score.
- $\alpha, \beta, \gamma, \delta, \eta$ = Dominion-selected weights.

The model should also flag bids that are suspiciously low. Bidding down contracts is useful only if the bids are realistic. Underbidding can create delays, change orders, quality issues, or safety risk.

7.4 Risk-adjusted bid selection

Dominion should compare vendors by risk-adjusted value, not price alone. One possible value function is:

$$V_{j,w}^{\text{risk}} = V_w - \text{Bid}_{j,w} - \rho_1 P_j^{\text{delay}} D_w - \rho_2 P_j^{\text{incident}} I_w - \rho_3 P_j^{\text{rework}} R_w$$

where:

- V_w = expected utility value of completing work package w .
- $\text{Bid}_{j,w}$ = contractor bid.
- P_j^{delay} = estimated probability of delay.
- D_w = cost of delay.
- P_j^{incident} = estimated probability of safety or compliance incident.
- I_w = cost of an incident.
- P_j^{rework} = estimated probability of rework.
- R_w = expected cost of rework.
- ρ_1, ρ_2, ρ_3 = risk weights set by Dominion.

The award rule could be:

$$j^* = \begin{cases} \text{manual review,} & \text{if no vendor meets qualification thresholds} \\ \arg \max_j V_{j,w}^{\text{risk}}, & \text{otherwise} \end{cases}$$

This keeps Dominion in control. The system recommends a shortlist, explains the tradeoffs, and flags risky bids. The market creates cost pressure, but Dominion keeps final judgment over safety, trust, and reliability.

7.5 Open-bidding workflow

The workflow should be simple:

1. GRID identifies a high-value DER, reliability, or infrastructure opportunity.
2. Dominion converts that opportunity into a standardized work package.
3. The platform checks which contractor qualifications are required.
4. Prequalified vendors are invited to bid.
5. Contractors submit cost, schedule, crew, safety, and technical information.
6. GRID scores bids across cost, risk, timing, geography, and credibility.
7. Dominion reviews the ranked shortlist.
8. Dominion awards the contract.
9. Completion outcomes update the vendor profile.

The feedback loop is the important part:

$$\text{Bid} \rightarrow \text{Award} \rightarrow \text{Execution} \rightarrow \text{Outcome} \rightarrow \text{Vendor score update}$$

Over time, the platform gives Dominion a procurement memory. Good contractors can prove quality with data. Contractors with repeated delays, rework, or safety issues become easier to detect. This makes bidding more useful because the system is not only comparing promises. It is comparing promises against past performance.

7.6 AI integration

AI should support the bidding process, not replace procurement judgment. The best use case is administrative and analytical support: drafting scopes, matching vendors, flagging unusual bids, and summarizing contractor histories.

Useful AI features could include:

- A work-package generator that turns GRID insights into draft bid scopes.
- A vendor-matching model based on geography, qualifications, and past work.
- Bid anomaly detection for unusually low, high, incomplete, or inconsistent bids.
- A safety-risk classifier based on incident history and job type.
- A schedule-risk model based on crew availability, season, location, and complexity.
- A document assistant for insurance, certifications, compliance forms, and prior project records.

A simple delay-risk model could be:

$$P_j^{\text{delay}} = \frac{1}{1 + e^{-z_j}}$$

$$z_j = \beta_0 + \beta_1 H_j + \beta_2 A_j + \beta_3 X_w + \beta_4 M_w + \beta_5 D_j$$

where H_j is historical delay rate, A_j is current availability, X_w is work complexity, M_w is seasonal or mobilization difficulty, and D_j is distance from the work site.

7.7 Simulation integration

Simulation should help Dominion compare procurement strategies before committing budget. For example, Dominion could compare the cheapest qualified vendor, the fastest vendor, the safest vendor, and the vendor with the highest risk-adjusted value.

A Monte Carlo simulation could estimate expected completion cost:

$$\text{TotalCost}_{j,w}^{(m)} = \text{Bid}_{j,w} + \text{DelayCost}_{j,w}^{(m)} + \text{ReworkCost}_{j,w}^{(m)} + \text{IncidentCost}_{j,w}^{(m)}$$

Across many simulation runs, Dominion could estimate:

- Expected total cost.
- Probability of finishing on time.
- Probability of exceeding budget.
- Expected customer impact.
- Expected grid value delivered by the completion date.
- Downside risk during peak season or storm season.

This gives the platform a planning role, not just a procurement role. Dominion could test whether it is better to split a large package across several vendors, award it to one proven contractor, or delay the work until crew capacity improves.

7.8 Behavioral economics integration

The contractor side has its own behavior problem. Vendors may avoid bidding if the process feels too relationship-dependent, paperwork-heavy, confusing, or unclear. Smaller qualified contractors may assume they cannot compete with established firms even when they meet the technical requirements.

The bidding layer should reduce that friction through:

- Clear qualification requirements.
- Standardized bid templates.
- Transparent scoring categories.
- Faster document upload and verification.
- Feedback when a bid is rejected.
- Contractor dashboards showing upcoming opportunities.
- Plain-language explanations of what Dominion values.

This does not mean Dominion should lower standards. It means the standards should be easier to understand. A clearer market can increase competition while still protecting safety, reliability, and credibility.

7.9 Scalability

A scalable version should connect six data layers:

Layer	Purpose
Customer intelligence	Identify adoption opportunities and barriers
Grid value model	Estimate peak reduction, avoided cost, and reliability value
Work package builder	Convert insights into contractable scopes
Vendor marketplace	Let prequalified contractors view and bid on work
Bid scoring engine	Compare bids by cost, risk, safety, geography, and timing
Execution tracker	Record completion, delays, incidents, and performance

This is stronger than a standalone DER dashboard. GRID would connect the full chain from insight to execution: customer targeting, grid-value modeling, standardized work definition, contractor competition, award review, and post-project learning.

7.10 Future optimization layer

Once customer targeting, utility-value modeling, and contractor bidding are connected, GRID can become an optimization platform. Dominion could choose which households or circuits to prioritize, which work packages to release, and which contractor portfolio best satisfies cost, reliability, safety, and equity constraints.

A simplified portfolio optimization problem is:

$$\max_{z_w, a_{j,w}} \sum_{w=1}^W z_w (V_w - \sum_{j=1}^J a_{j,w} \text{RiskAdjustedCost}_{j,w})$$

subject to:

$$\sum_{w=1}^W \sum_{j=1}^J z_w a_{j,w} \text{Bid}_{j,w} \leq B$$

$$\sum_{j=1}^J a_{j,w} = 1 \quad \text{for each awarded work package } w$$

$$S_{j,w}^{\text{contract}} \geq S_{\min}$$

where z_w indicates whether work package w is released, $a_{j,w}$ indicates whether contractor j is assigned to work package w , V_w is the expected utility value, and B is the procurement budget.

This optimizer would let Dominion answer a higher-level question: with limited budget, limited crew capacity, and reliability constraints, which combination of DER campaigns, infrastructure work, and contractor awards creates the most value?

7.11 Final future direction

The strongest version of the idea is not “open bidding for utilities.” That is too broad and too risky. The stronger version is a controlled bidding layer for standardized DER work.

GRID should identify where DER value exists, translate that value into clear work packages, invite qualified contractors to compete, compare bids with cost and trust metrics, and learn from the work after it is completed. That framing validates the idea of bidding down contracts without pretending utility work is a simple commodity.

The long-term vision is:

Find the right household, diagnose the right barrier, estimate the grid value, define the work clearly, invite qualified contractors, select the best risk-adjusted bid, and learn from the result.